

Figure 0.1 · Proposition 1 Mechanics: Serial Write Arbitration

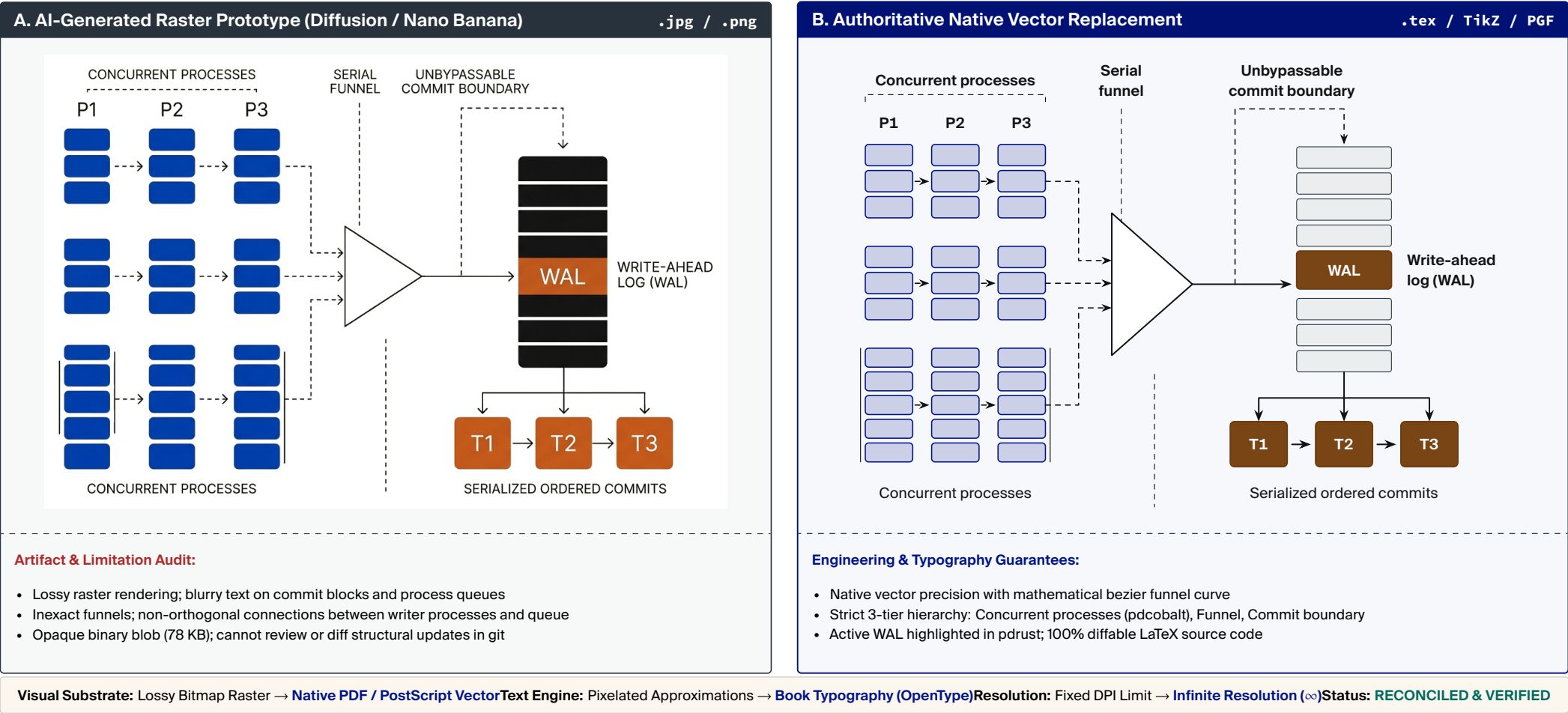


Figure 0.2 · Proposition 2 Mechanics: Monotonic Capability Attenuation

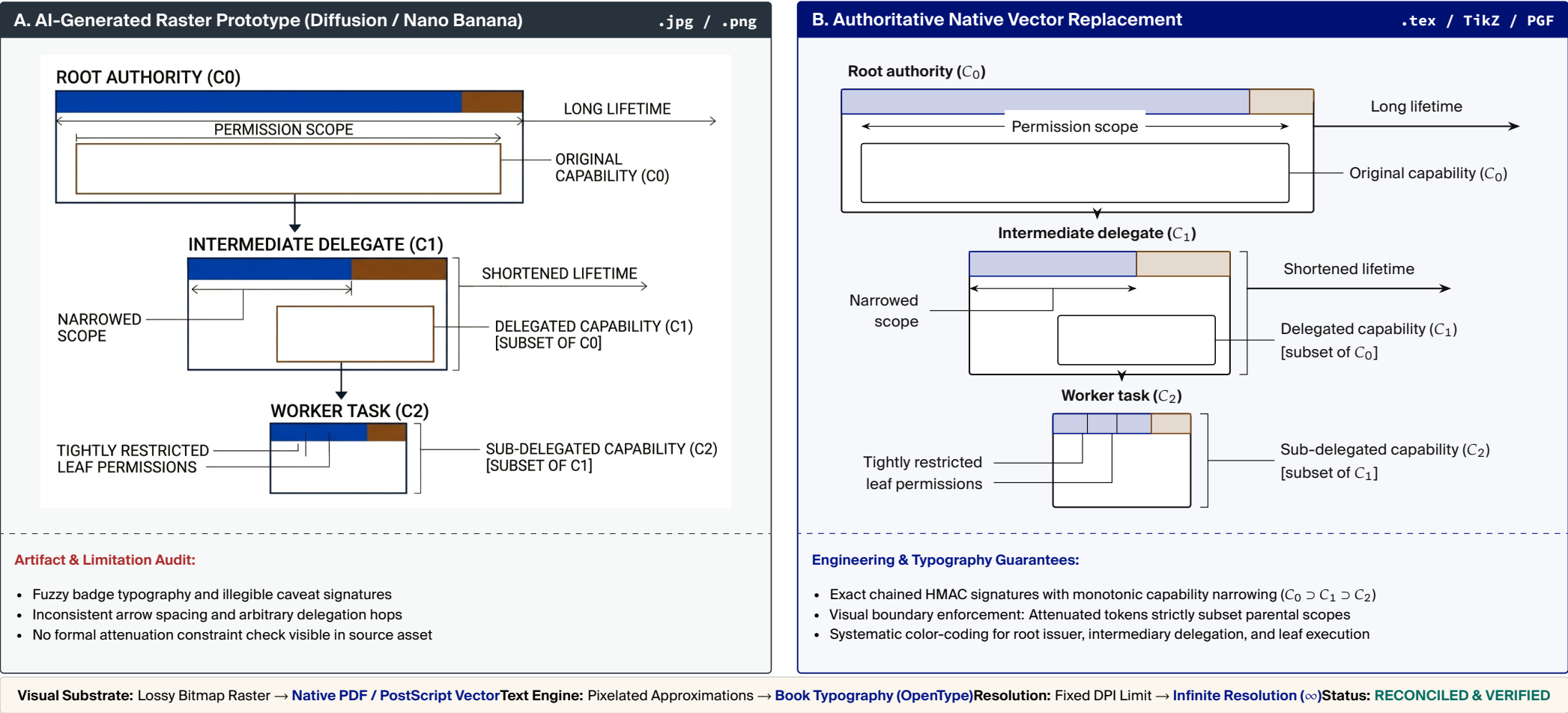
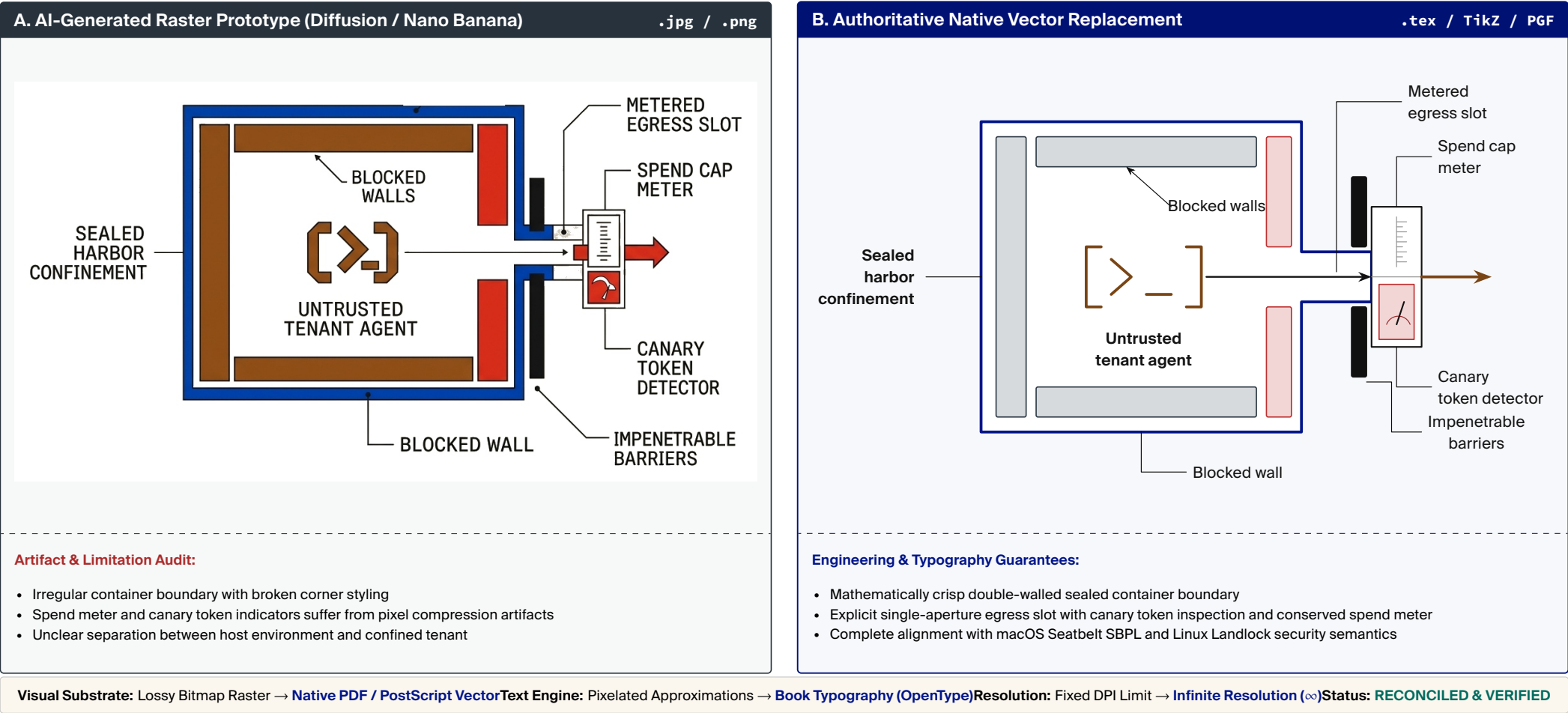


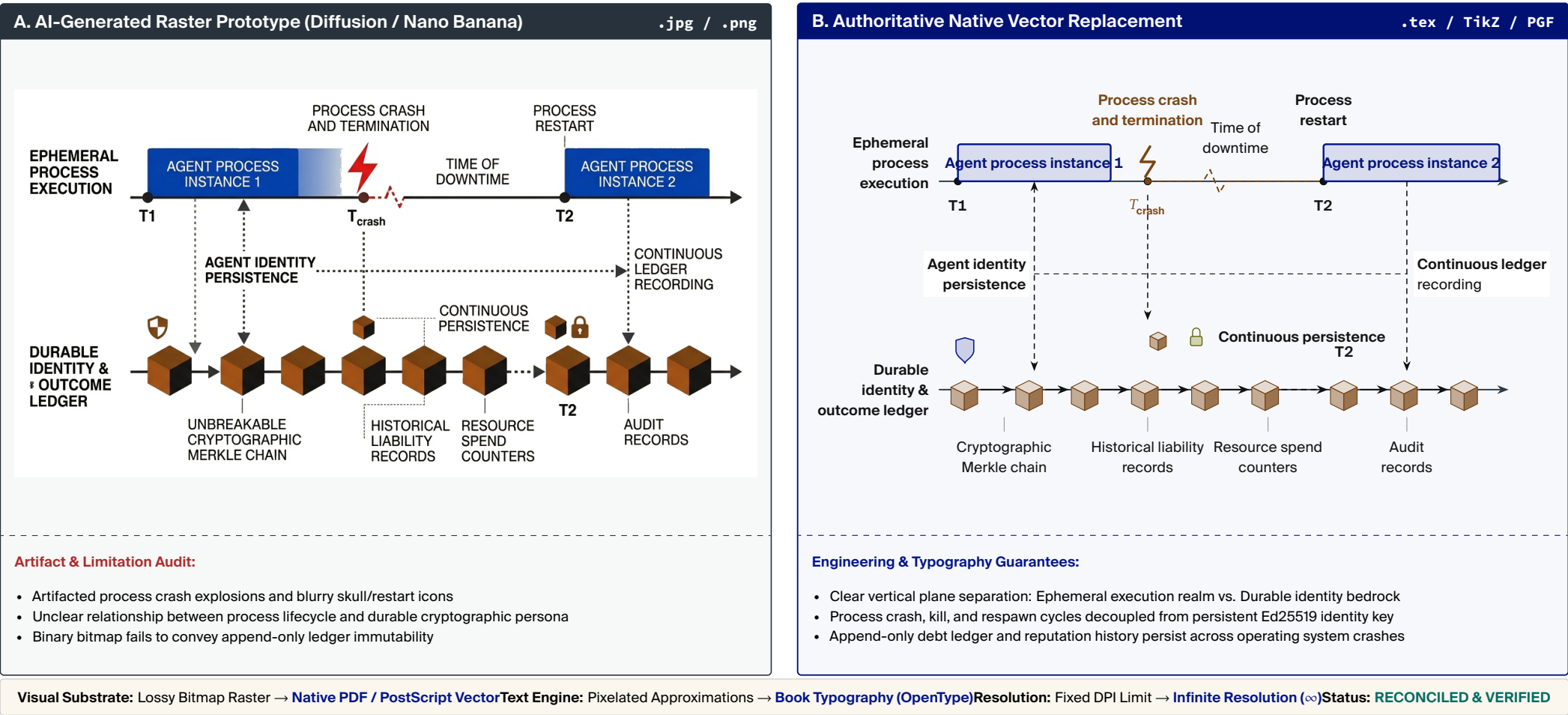
Figure 0.3 · Proposition 3 Mechanics: Sealed Room Confinement



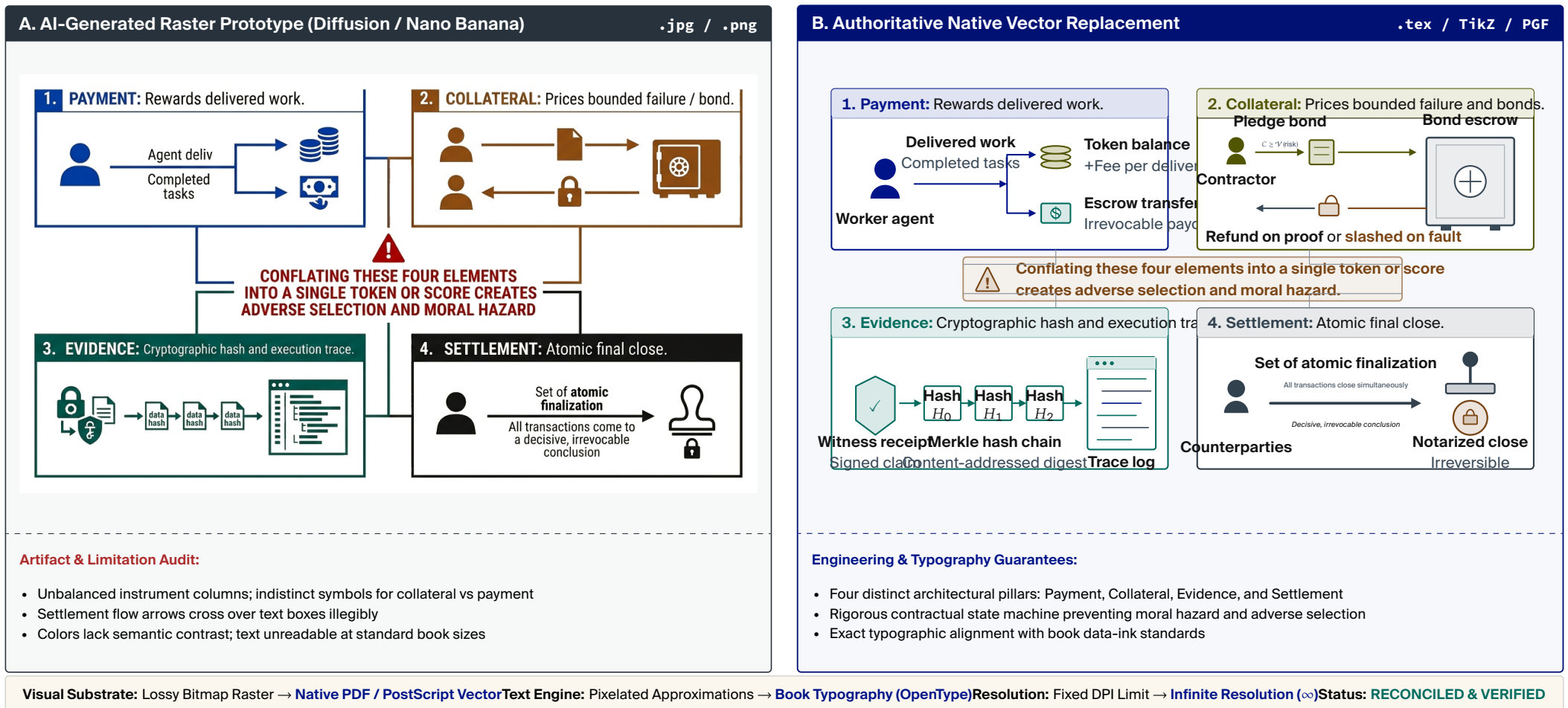
Section 0.2.4 · Supervisory Systems · Attention Budget & Drilling



Figure 0.5 - Proposition 5 Mechanics: Durable Persona vs. Ephemeral Process



Section 0.2.6 · Agent Economics · Market Instruments & Settlement



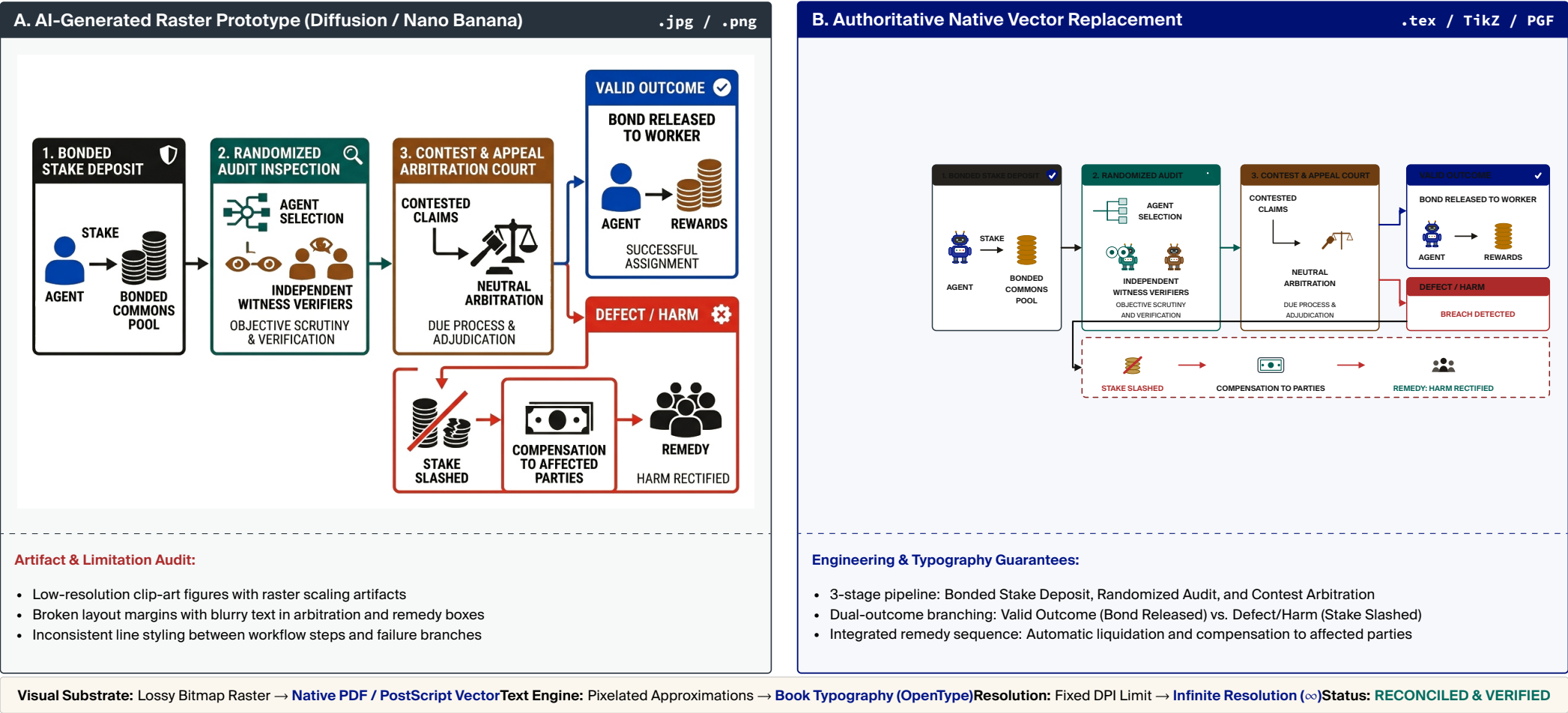


Figure 0.8 · Proposition 8 Mechanics: Cross-Harbor Relay & Zero-Trust Federation Section 0.2.8 · Distributed Architecture · Relay Fabric & Proof Gossip

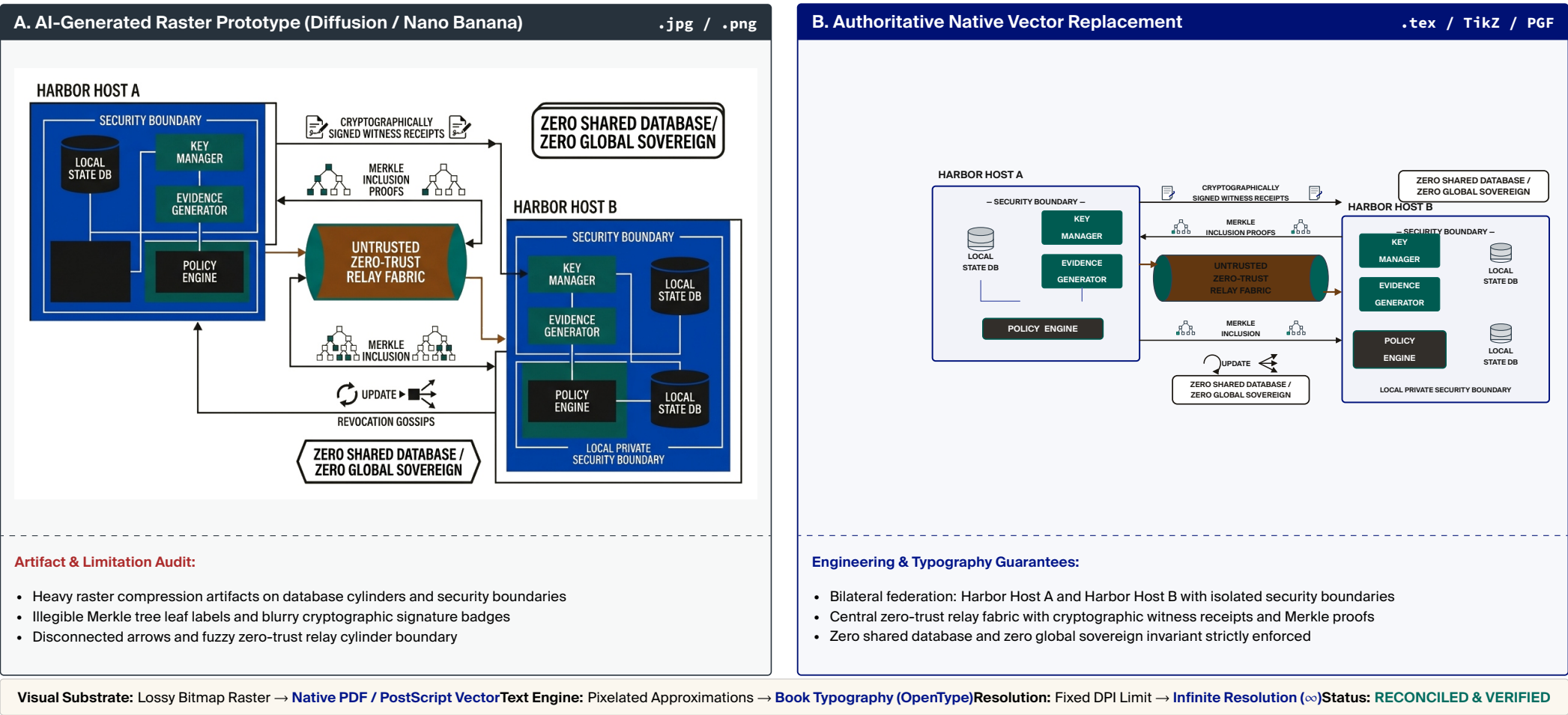
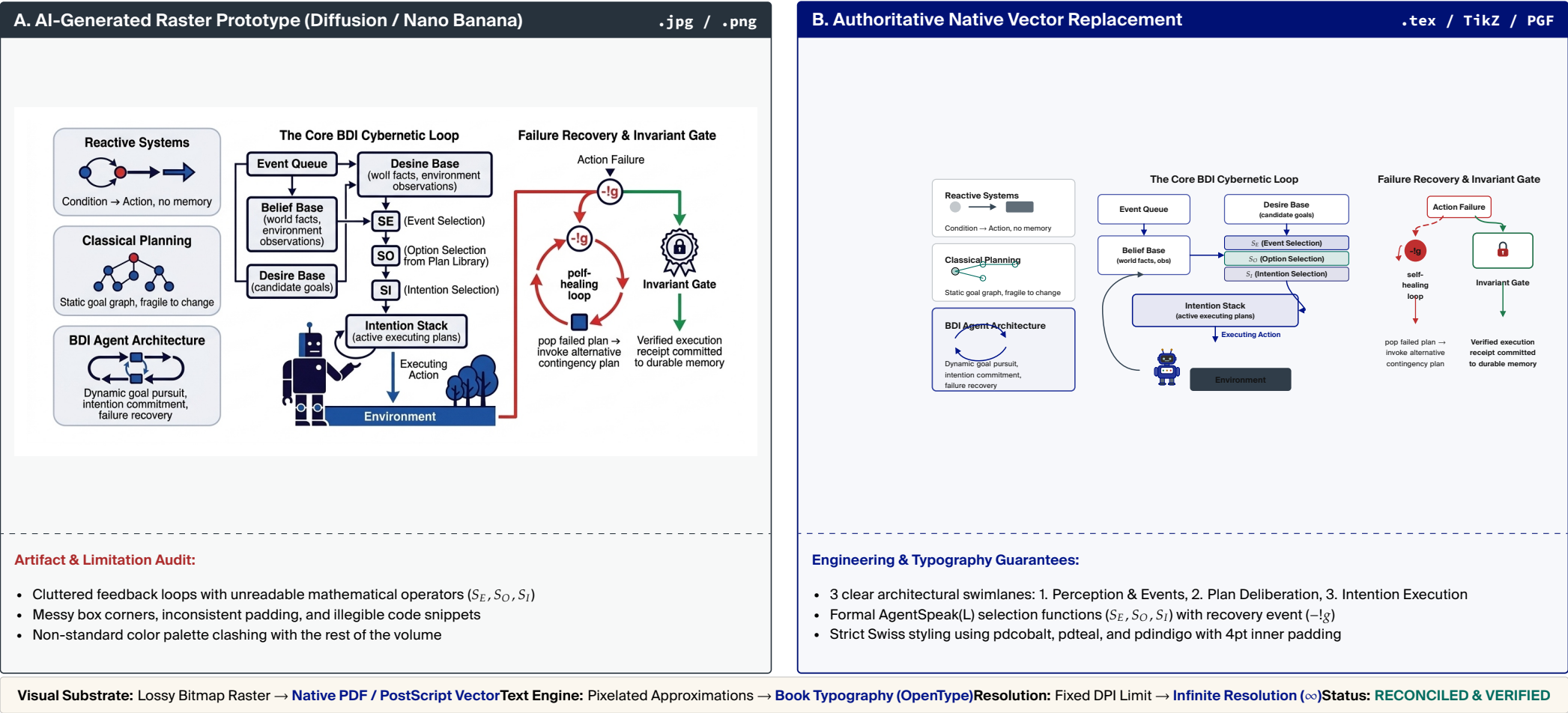
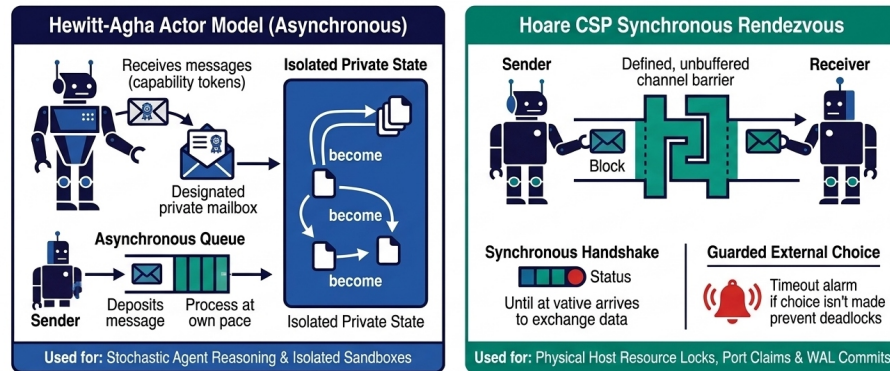


Figure 0.9 · BDI Execution Flow and Operational Choice Architecture



A. AI-Generated Raster Prototype (Diffusion / Nano Banana)

.jpg / .png



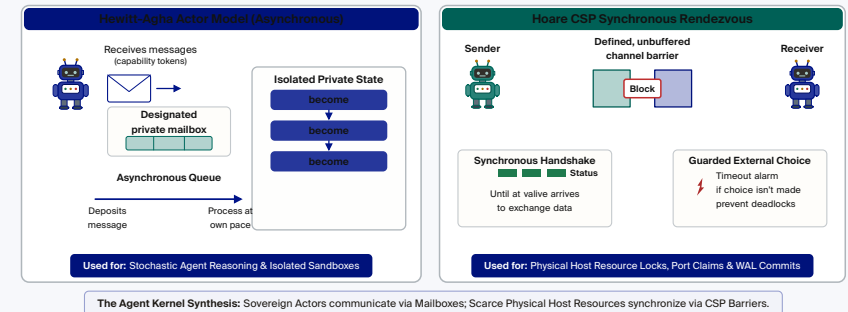
The Agent Kernel Synthesis: Sovereign Actors communicate via Mailboxes; Scarce Physical Host Resources synchronize via CSP Barriers.

Artifact & Limitation Audit:

- Asymmetric lifelines with wavy paths and distorted mailbox queues
- Rendezvous barrier rendered as a fuzzy blob rather than an instantaneous synchronization
- Lack of formal notation for actor replacement behaviors (become)

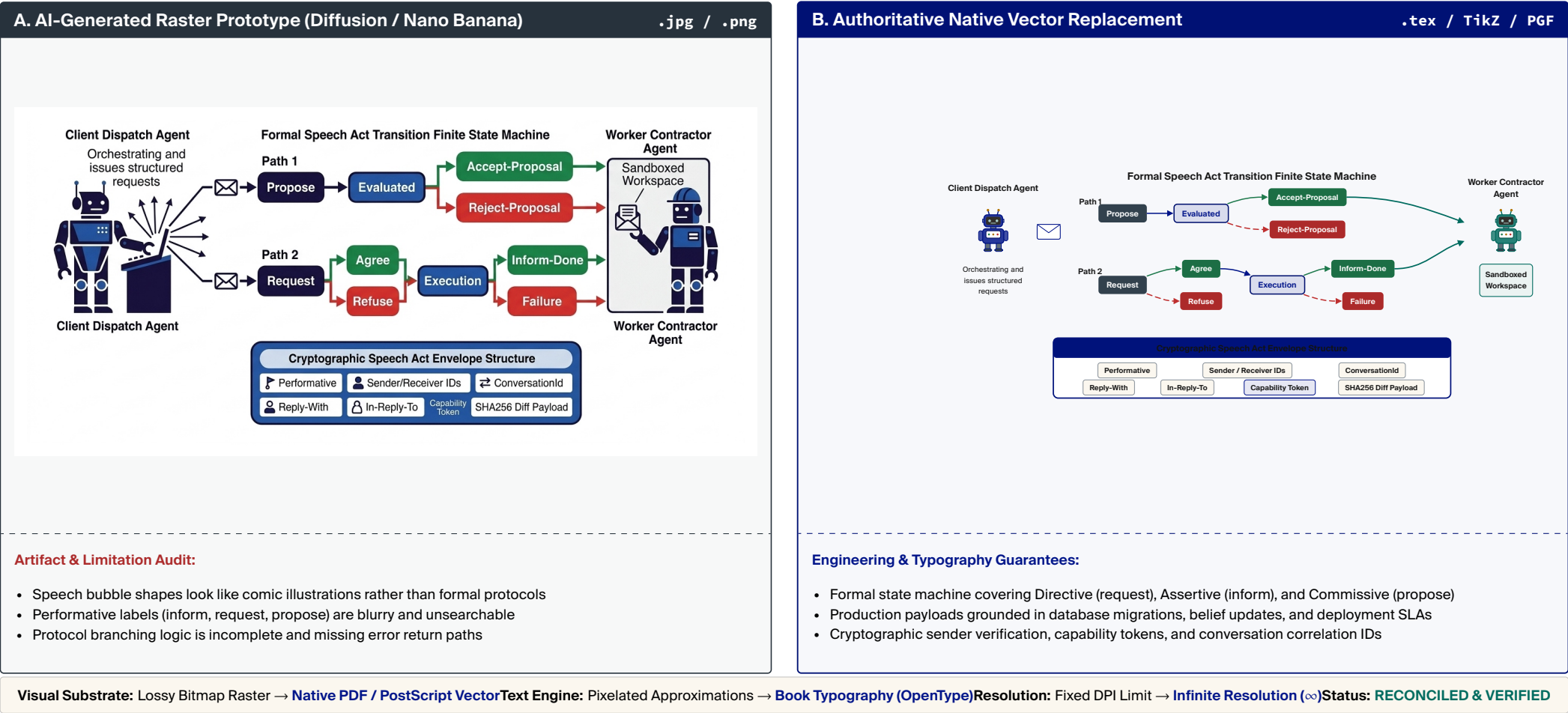
B. Authoritative Native Vector Replacement

.tex / TikZ / PGF



Engineering & Typography Guarantees:

- Side-by-side comparison of asynchronous Actor mailboxes vs. synchronous CSP rendezvous
- Explicit zero-duration synchronization barrier ($\Delta t = 0$) with lease timeout guards
- Formal state transitions: $\text{become}(B')$, unbuffered channels, and external choice



Section 0.1.6 · Task Allocation & Market Protocols · Smith (1980) CNP

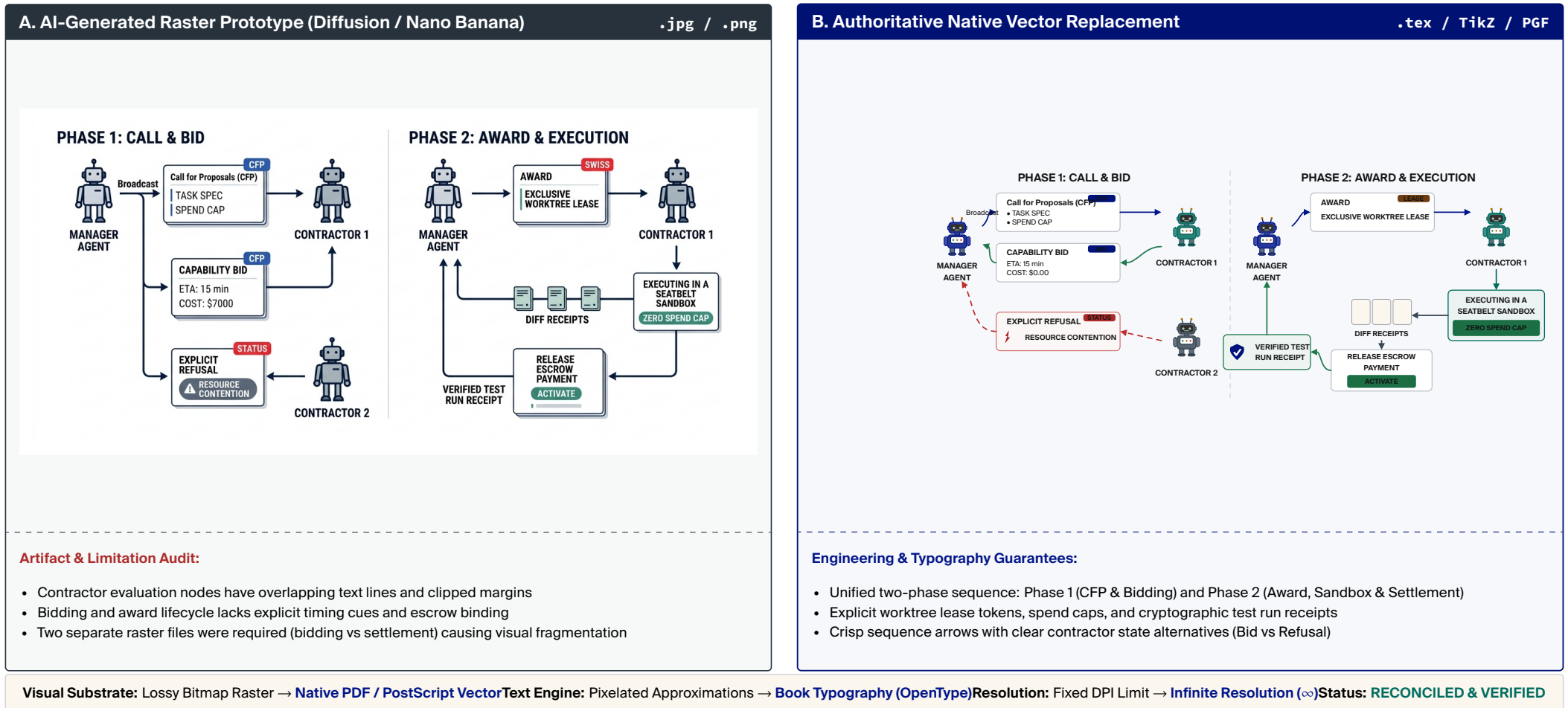
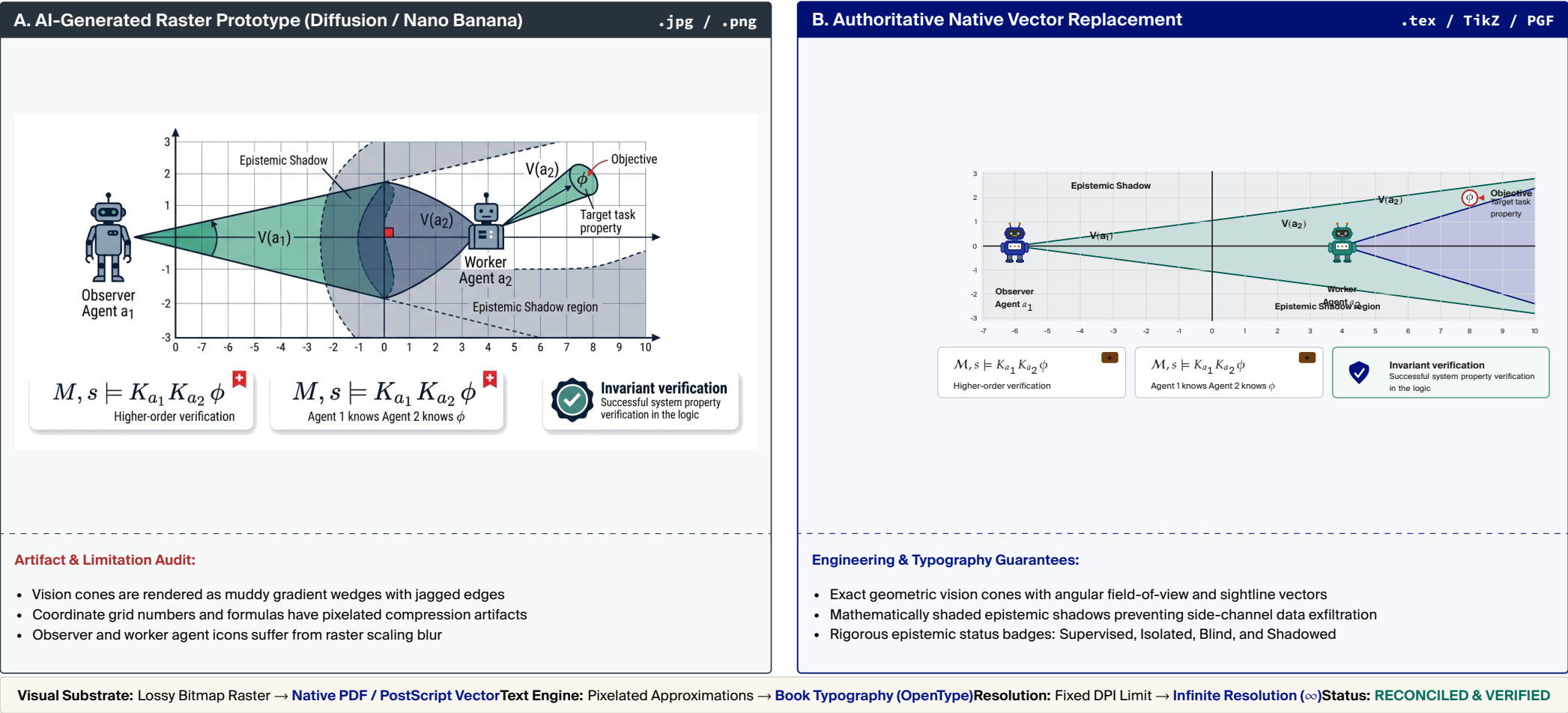


Figure 0.13 · Dynamic Epistemic Observation Cones in Big Brother Logic



Section 0.1.8 · Social Choice & Game Theory · Arrow (1951)

.jpg / .png



- .tex / TikZ / PGF

Section 0.3.3 · Tool Calling & Extensibility · Skill Architecture

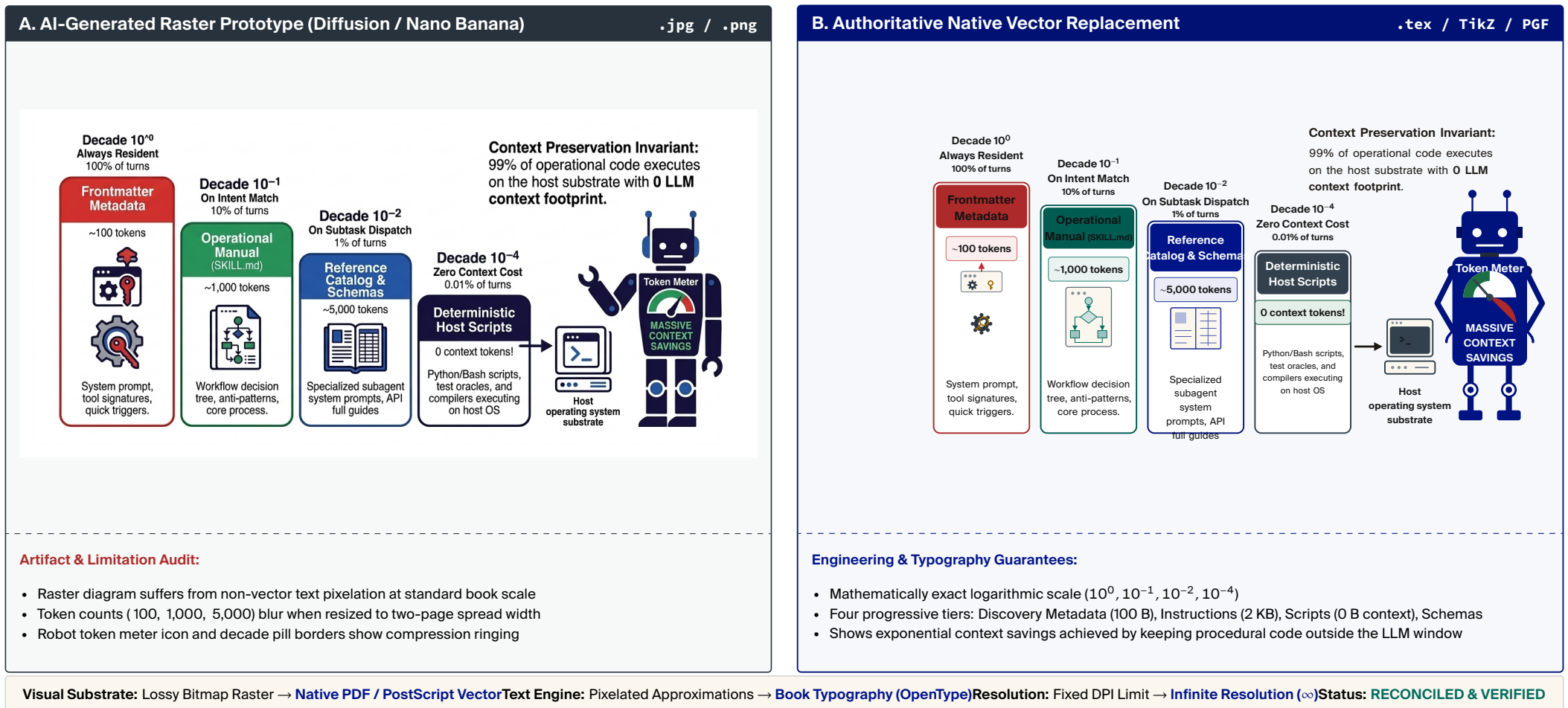
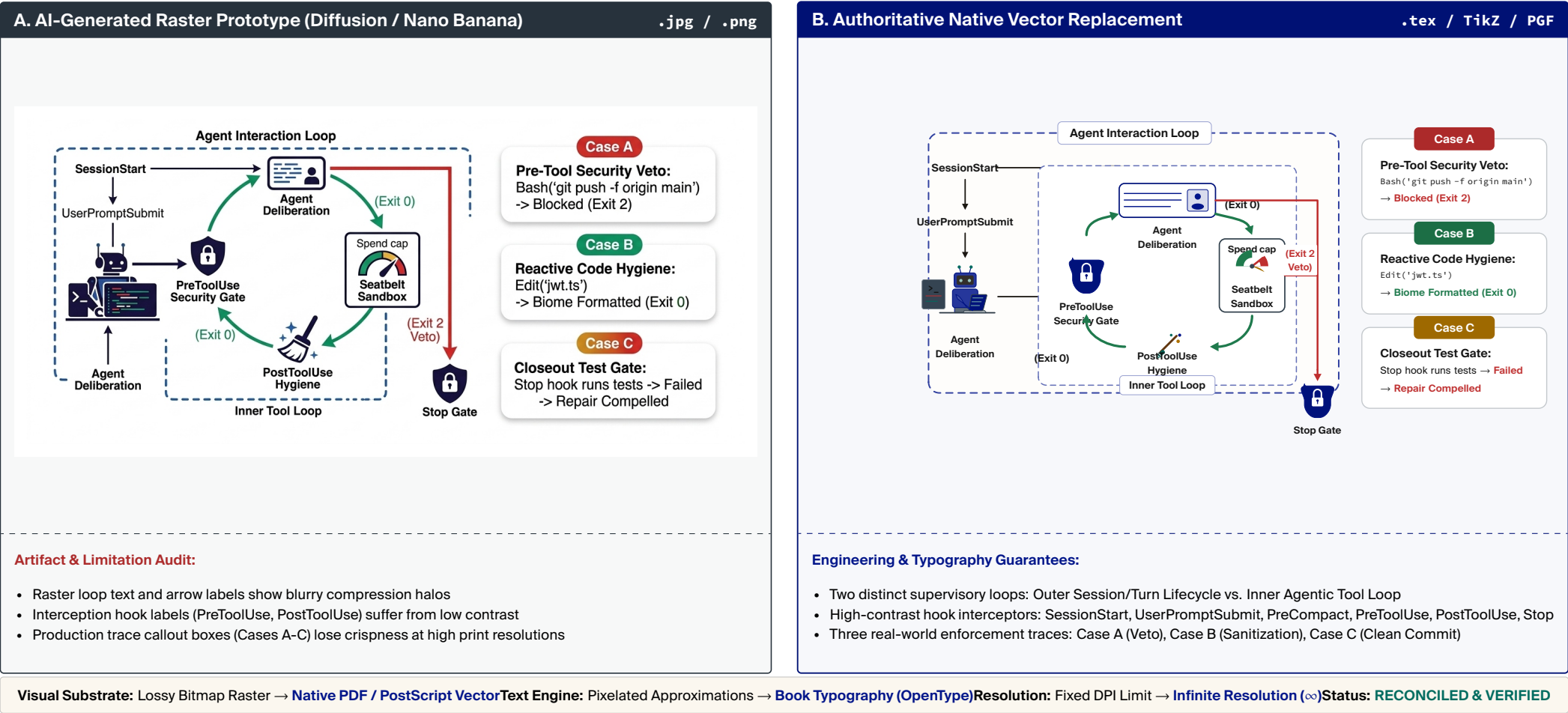


Figure 0.16 · The Agent Lifecycle Hook Cross-Section



Section 0.3.5 · Supervisory Control Theory · Discrete-Event Systems

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- Automata state nodes are distorted ellipses with unreadable transition labels
- Controllable vs uncontrollable event sets (Σ_c vs Σ_u) are visually indistinguishable
- Mathematical notation for supervisory feedback control is blurry

.tex / TikZ / PGF



- Formal Moore-automata state machine with controllable events (Σ_c) and uncontrollable events (Σ_u)
- Closed-loop supervisor S dynamically disabling unsafe transitions to enforce language $K \subset L(G)$
- Direct application to sandboxed agent tool execution and mutual exclusion invariants

Visual Substrate: Lossy Bitmap Raster → **Native PDF / PostScript VectorText Engine:** Pixelated Approximations → **Book Typography (OpenType)**Resolution: Fixed DPI Limit → **Infinite Resolution (∞)**Status: **RECONCILED & VERIFIED**

Section 0.4.1 · Closed-Loop Governance · Three-Tier Operating Architecture